
Assignment: The Twin Detector — File Edition

Problem Statement

The school's admission office has handed you a file students.txt containing student records, one per line in the format:

name,rollNo,age

Your job is to read the file, create a Student object for each line, store them in a fixed-size array, and find all twin pairs — students who share the same roll number AND age.

Part 1 — The Student Class

Create a class Student with:

1. Three private fields: name (String), rollNo (int), age (int)
2. A parameterized constructor Student(String name, int rollNo, int age)
3. Public getters for all three fields
4. A method boolean hasSameAge(Student other) that returns true if both students have the same age
5. A method boolean isTwin(Student other) that returns true if both students have the same rollNo AND age
    - Important: isTwin() must reuse hasSameAge() internally — do not write the age check twice

Part 2 — Read the File into an Array

In your main method:

1. Create a Student array of size 500 (we'll assume no more than 500 students)
2. Read students.txt line by line using a Scanner
3. For each line, split on "," to get name, rollNo, and age and parse numbers with Integer.parseInt()
4. Create a Student using the parameterized constructor and store it in the array
5. Keep an int count variable to track how many students you actually loaded

Part 3 — Find Twin Pairs

1. Use a nested loop with j = i + 1 to compare every pair of students
2. If students[i].isTwin(students[j]) is true, print them as a twin pair and increment a counter

3. Print the total number of twin pairs at the end

Expected Output Format

Loaded 300 students.

=== Twin Pairs Found ===
Pair 1: <name1>, <name2> (Roll: X, Age: Y)
Pair 2: ...
...

Total twin pairs: K

Hints

- Scanner sc = new Scanner(new File("students.txt")); — add throws Exception on main
- Skip empty lines with: if (line.isBlank()) continue;
- Access fields only via getters — since fields are private
- Always loop up to count, never up to students.length (the empty slots will cause NullPointerException)

Submission Rules

1. File name must be: YourName_TwinDetector.java
2. Place it inside the folder: assignments/getters-setters/
3. Include your students.txt file in the same folder
4. Commit message format:
   Add Twin Detector (File Edition) - <Your Full Name>
5. Push to the repository before the deadline

What This Assignment Tests

- Writing and using getters and setters (encapsulation)
- Method reuse — isTwin() calling hasSameAge()
- File reading with Scanner
- String parsing with split() and Integer.parseInt()
- Working with fixed-size arrays and a count variable
- Nested loops to compare every pair